This is a change of paradigm!

(1) THE CAUSE OF EMERGING MUTATIONS: PROTEIN DAMAGE MATTERS MORE THAN DNA DAMAGE

(2) *THE ORIGIN OF EMERGING MUTATIONS: PROTEIN DAMAGE IS MORE MUTAGENIC THAN DNA DAMAGE*

(3) PROTEIN DAMAGE IS THE LEADING CAUSE OF EMERGING MUTATIONS

(4) PROTEIN DAMAGE DETERMINES MUTATION RATES

(5) PROTEIN DAMAGE IS THE PRINCIPAL CAUSE OF MUTATIONS

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Evolution of biological diversity and emergence of many diseases, prominently cancer, are the consequences of inherited and/or somatically acquired mutations. Correcting disease-causing heritable mutations and reducing ongoing somatic mutation load is of relevance to gene therapy and disease prevention, respectively. Evidently, new mutations arise as the consequence of DNA copy errors and a variety of DNA damages. But, this mechanistic statement does not identify the root cause of mutations, i.e., what determines their occurrence at given frequencies? Identifying the bottlenecks in reduction of spontaneous mutation rates and frequencies of mutations induced by sunlight, ionizing radiation and genotoxic chemicals, would identify the root cause of mutations and define targets for reducing mutation rates.
With this in mind, extensive screenings were undertaken with T4 phage and bacterium *Escherichia coli* to find mutations whose phenotype is substantial reduction in spontaneous mutation rates, but did not reveal the existence of any such genuine “anti-mutator” mutants. The explanation could be that DNA replication errors and DNA damage contribute about equally to spontaneous mutation rates. Thus, single mutations reducing only one source of emerging mutations could decrease mutation rate maximally by 50% and simultaneous double mutants would be extremely rare.

Using the genetic tools of the best explored organism in mutation research, the bacterium *Escherichia coli*, we found a physiological solution to this conundrum by discovering that oxidative protein damage determines mutation rates and acts as the bottleneck to the emergence of spontaneous and induced mutations. These findings are paradigm-changing from DNA-centric to protein-centric view on mutations – a conceptual change of relevance to biomedical research on aging and age-related diseases.

The fact that mutations reside in DNA does not mean that either DNA by itself, or damage inflicted to DNA, determine mutation rates. The destiny of DNA, including its mutation rates, is governed by the activities of the DNA proteome – proteins involved in DNA transactions.
For instance, very high radiation doses are not significantly mutagenic to the extremely resistant bacterium, *Deinococcus radiodurans*, endowed with extremely efficient anti-oxidant protection of proteins, but not of DNA. Moreover, mutants of standard bacteria with inactive SOS response generally do not mutate by DNA damaging agents. Hence, emerging mutations are not unavoidable stochastic chemical accidents, but the results of altered activity and/or fidelity of proteins that replicate and repair DNA. These properties of the DNA proteome are negatively affected by oxidative protein damage accumulating in all cells and by particular rare "mutator" mutations.

The key findings are:

(i) Physiological spontaneous mutation rates can be reduced in *E. coli* by at least an order of magnitude by reducing, selectively, the oxidative damage to the proteome at constant physiological levels of reactive oxygen species (ROS).

(ii) Reducing protein oxidation either by increasing intrinsic resistance of proteins to oxidative damage or by reducing ROS levels (by Trolox, a water-soluble vitamin E) leads to similar reduction of mutation rate in *E. coli*.

(iii) Oxidative damage inflicted selectively to the proteome proved to be much more mutagenic than exclusive DNA damage: mutations arise with approximately 7th power of total cellular protein carbonylation (an un-reparable oxidative protein damage) while usually increasing linearly with inflicted DNA damage.

(iv) Both, spontaneous and UV light-induced mutations increase with protein carbonylation with similar slope.

(v) Proteins have acquired evolved structures that minimize the susceptibility to oxidative damage. However, that resistance is fragile: even subtle reversible misfolding predisposes proteins to oxidative damage that fixes the misfolded state and only then – upon oxidation of the DNA proteome - causes the emergence of mutations. Misfolding alone, without oxidative damage, appears reversible by chaperone activities and has at best a minor effect on mutation rate.

(vi) Subtoxic concentrations of streptomycin, an antibiotic which increaes ribosomal misreading, cause rapid accumulation of carbonylated proteins accompanied by increased mutation rates under aerobic but not under anaerobic conditions (show!).

(vii) Our results lend experimental support to L. Orgel's error catastrophe theory (i.e., bad proteins make bad DNA leading to even worst proteins, and so on…) of aging modifying it by an essential role of oxidative modification of proteins.

<u>Discussion points:</u>

High mutagenicity of proteomic, mainly oxidative, damage makes sense. To make nearly perfect DNA copies, (a) the pools of four dNTPs have to be synthesized in optimal concentration ratios corresponding to the ratio of the four bases in the replicating genome, (b) dNTP pools must be readily sanitized by specific hydrolases to eliminate, among others, dUTP

(by dUTPase) and 8-oxo-dGTP (by MutT protein) that are mutagenic when incorporated into DNA, (c) before replication, DNA templates must also be sanitized from mutagenic modifications by several DNA repair systems, (d) protein complex of the DNA replication machinery must achieve high fidelity of DNA synthesis, then (e) correct, in a short time window, over 99% of emerging replication errors; therefore the complex of mismatch repair proteins should work with top efficacy and precision; and finally, (f) a plethora of DNA damage repair and DNA damage signalling protein complexes must perfectly interact and precisely catalyse numerous reactions to maintain a nearly perfect DNA. Protein damage can affect all six fidelity steps engaging together a significant fraction of the proteome (at least abouut one hundred proteins).
In addition, physiological post-synthetic protein modifications (PTM) and specific protein complexes are involved in fine-tuning of nearly all cellular processes – including DNA replication and repair - and require highly specific protein interactions that are perturbed by protein damage (ref). Together with (a) through (f) steps, this constitutes a large proteomic multi-target for oxidative damage with its functional and toxic consequences. Hence, damage to DNA replication, repair and PTM proteomes appears to set the lower limit to mutation rates in coding DNA sequences (exomes), that are similar across the kingdom of life (around 10^{-9} mutations per replicated base). For *E. coli*, the extrapolation of proteomic oxidative protein carbonylation to zero predicts lowest error rate per replicated base to about $7x10^{-11}$, i.e., the lowest intrinsic mutation rate in replication of *E. coli* genome under aerobic conditions.

Mechanistically, new mutations emerge from residual copy errors (mismatched base pairs) and residual mutagenic DNA modifications that have escaped biosynthetic proofreading and DNA mismatch and damage repair mechanisms executed by dedicated proteins. Therefore, one could predict that – similarly to inherited mutations - damage to any and all of such dedicated proteins should be more mutagenic than the inflicted DNA damage because the latter is mitigated by the high efficacy and precision of an intact DNA proteome.

Here, this prediction is positively tested and extends into the concept of ribosomal error catastrophe, in particular when oxidative damage to ribosomes, similarly to ribosome-binding drugs, leads to the synthesis of erroneous misfolded proteins prone to oxidation. Such proteomic error catastrophe generates high mutation rates that are just one (“mutator”) among many phenotypes of the general cellular error/oxidation catastrophe.
The well-documented quantitatively similar age-related accumulation of carbonylated proteins across the lifespans of species (from nematode to human) now argues that oxidative protein damage is also the likely root cause of aging and of age-related diseases. Genetic, epigenetic and direct functional alterations in the course of aging can all be the consequences of age-related accumulation of damaged proteins and account for all particular phenotypes and “biomarkers” of aging and age-related diseases. Conceptually, this is a modified L. Orgels’s error catastrophe theory complemented by introducing the key role of oxidative damage to proteins.

Gist:
Protein damage is the leading cause of emerging spontaneous and induced mutations, via reduced fidelity of DNA synthesis and reduced efficacy of DNA repair systems, respectively.
(Compare mutation rates in mutH mutant and mesure oxid damage in DNA- done!)

OTHER CONSIDERATIONS

High mutation rates – non-self antigens - tissue cleaning by MHC.
Drake's rule: correct normalisation by exome (not genome) size! – get universal spont. mutation rates per exome!
(exome sizes are similar from nematode to man!!!).

I - COLLABORATION WITH ROGULJA:
Get the guts from young, mid-life and dying "no sleep" and "normal sleep" mice: quantitate and identify the carbonylated proteins

II - COLLABORATION WITH THE ETH TEAM:

1. Make hi-fi ribosomal mutant mice. Is there extension of life span +/- rapamycin?
2. Add antioxidants in drinking water of WT and ribosomal ram and hi-fi mutant mice. Extension of lifespan? How much?
3. Cross ram mutant mice with chaperone deletion mutants to get double mutants. Exaggerated results? Antioxidant effects?
4. Cross hi-fi ribosomal mutants with chaperon deletion mutants to get double mutants.
5. Predicting specific diseases by knock-out of particular chaperone genes. Relate diseases to the client proteins of deleted chaperones!!! (see mouse chaperone mutants from Lindquist and Morimoto collection). Effect of antioxidants in drinking water on disease onset?

III – MORE WORK FOR US TO GET DATA FOR MAMMALIAN CELLS ?:

1. Test mutagenicity (6-TG and Ouabain resistance) of subtoxic concentrations ricin, abrin (ribosomal drugs) and geldanamycin (anti-chaperon drug) +/- Trolox.
2. Measure protein carbonylation in #1